Exp No : 07
24/09/2026

Sliding window Protocol

Aim: write a program to implement flow control at data link using sliding window protocol.

Program:

```
import time
message = input ("enter the text message : ")
window_size = int (input ("enter the window size : "))
error_frame = int (input ("enter frame : "))
if window_size <= 0:
        print ("Invalid window size")
        exit()
Print ("sliding window Protocol")
Print ("window size :", window_size)
start = 0
expected_frame = 1
error_induced = false
while start < len (message):
    end = min (start + window_size, len (message))
    Print ("Sender buffer")
    Sender_buffer = []
for i in range (start, end):
        frame_no = i+1
        data = message [i]
        sends_buffer.append ([frame_no, data])
```

```
Print("Frame_no =", frame_no, "Data =", data)
Print("\n Sending frames:..")
time.sleep(1)
Print("Receiver")
error_formed = false
for frame in Sender_Buffer:
        frame_no = frame[0]
        data = frame[:]
if frame_no == error_frame and not error_induced:
        received_frame = frame_no + 1
        error_induced = True
else:
        received_frame = frame_no
Print("Received Frame :", received_frame, "Data:", data)
if received_frame == expected_frame:
        Print("Frame", received_frame, "is Correct")
        expected_frame += 1
else:
        Print("\n Error Detected")
        Print("Expected Frame :", expected_frame)
        Print("Received frame :", received_frame)
Print("Receiver Buffer")
Print("Nack =", expected_frame)
```

```
Print("\n Nack received by sender")
Print("Resending frames from:", expected_frame)
    error_found = true
    break
if error_found:
        start = expected_frame - 1
        time.sleep(1)
        Continue
    Print("Receiver Buffer")
    ack = expected_frame
    Print("Ack =", ack)
    Print("\n Ack received by sender")
    Print("expected Ack:", end+1)
        if ack == end+1:
            Print("\n Ack is Correct")
            Print("sliding window moves forward")
            Start=end
else:
    Print("\n Incorrect Ack")
    Print("waiting for Correct Ack")
Print("Transmission Completed")
Print("\n Original message:", message)
Print("All frames successfully transmitted")
```

output:

enter the test message : hello

Enter the window size = 3

message : hello

window size : 3

Sender buffer

Frame_no = 1 Data = h

frame_no = 2 Data = e

frame_no = 3 Data = l

Receiver

Received frame : 1 Data : h

Frame 1 is Correct

Received frame : 2 Data : e

Frame 2 is Correct

Received frame : 8 Data : l

Error Detected

expected frame : 3

Received frame : 8

Receiver buffer

Nack = 3

Nack. received by sender

Resending frames from frames 3.

Sender buffer

frame_no = 3 Data = l

frame_no = 4 Data = l

frame_no = 5 Data = 0

Sending frames..

Receiver

Received frame : 3 Data : l

Frame 3 is Correct

Received frame : 4 Data : l

Frame 4 is Correct

Received 5 is Correct

Receiver buffer

Ack = 6

Ack received by Sender

expected Ack : 6

Ack is Correct

Sliding window move forward

transmission Completed

original message : Hello

All frames successfully transmitted

Result:

Thus flow control at data clink using sliding window Protocol is done successfully.